vibrational contributions to the energy of a string depend on the size of the circular dimension. For this purpose, it proves convenient to separate the vibrational motion of strings into two categories: uniform and ordinary vibrations. Ordinary vibrations refer to the usual oscillations we have discussed repeatedly, such as those illustrated in Figure 6.2; uniform vibrations refer to even simpler motion: the overall motion of string as it slides from one position to another without changing its shape. All string motion is a combination of sliding and oscillating—of uniform and ordinary vibrations—but for the present discussion it is easier to separate them in this manner. In fact, the ordinary vibrations will not play a central part in our reasoning, and we will therefore include their effects only after we have finished giving the gist of the argument.

Here are the two essential observations. First, uniform vibrational excitations of a string have energies that are inversely proportional to the radius of the circular dimension. This is a direct consequence of the quantum-mechanical uncertainty principle: A smaller radius more strictly confines a string and therefore, through quantum-mechanical claustrophobia, increases the amount of energy in its motion. So, as the radius of the circular dimension decreases, the energy of motion of the string necessarily increases—the hallmark feature of an inverse proportionality. Second, as found in the preceding section, the winding mode energies are directly—not inversely—proportional to the radius. Remember, this is because the minimum length of wound strings, and hence their minimum energy, is proportional to the radius. These two observations establish that large values of the radius imply large winding energies and small vibration energies, whereas small values of the radius imply small winding energies and large vibration energies.

This leads us to the key fact: For any large circular radius of the Garden-hose universe, there is a corresponding small circular radius for which the winding energies of strings in the former universe equal the vibration energies of strings in the latter, and vibration energies of strings in the former equal winding energies of strings in the latter. As physical properties are sensitive to the total energy of a string configuration—and not to how the energy is divided between vibration and winding contributions—there is no physical distinction between these geometrically distinct forms for the Garden-hose universe. And